

MICRO COMPUTER INSTITUTE

Advanced Diploma in Computer Applications

Foundation Study Pack - Volume 1
ADCA | Duration: 12 Months

Learn - Practise - Apply

A structured bilingual-support handbook for classroom, lab and online learners.

Student use: Read each module, complete the lab activity, save evidence in your portfolio, and discuss doubts with your instructor.

Hindi support (Roman): Har module ko padhen, practical lab activity poori karen, apna work save karen aur doubt teacher se poochhen.

Course Roadmap

This 12 Months programme is organised into 10 learning modules. The sequence may be adjusted by the instructor to match the batch calendar.

No.	Module	Learning focus
1	Computer Fundamentals	Hardware, software, operating systems, files, folders and safe computer use.
2	Office Productivity	Word documents, Excel calculations, PowerPoint presentations and printing.
3	Advanced Excel & MIS	Functions, lookup formulas, PivotTables, validation, dashboards and reports.
4	Database Concepts	Tables, records, keys, queries, forms, reports and data backup.
5	Accounting with Tally	Company creation, ledgers, vouchers, inventory, GST and reports.
6	DTP & Graphics	Page layout, typography, image preparation and print-ready documents.
7	Web Fundamentals	HTML, CSS, responsive pages, domains, hosting and publishing.
8	Programming Logic	Algorithms, flowcharts, variables, conditions, loops and functions.
9	Internet & Cyber Safety	Email, cloud tools, privacy, passwords, phishing and digital citizenship.
10	Capstone Project	Plan, build, test, document and present an office automation solution.

Module 1: Computer Fundamentals

Hardware, software, operating systems, files, folders and safe computer use.

Key ideas

- Explain the purpose and essential vocabulary of computer fundamentals.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates computer fundamentals. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 2: Office Productivity

Word documents, Excel calculations, PowerPoint presentations and printing.

Key ideas

- Explain the purpose and essential vocabulary of office productivity.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates office productivity. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 3: Advanced Excel & MIS

Functions, lookup formulas, PivotTables, validation, dashboards and reports.

Key ideas

- Explain the purpose and essential vocabulary of advanced excel & mis.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates advanced excel & mis. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 4: Database Concepts

Tables, records, keys, queries, forms, reports and data backup.

Key ideas

- Explain the purpose and essential vocabulary of database concepts.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates database concepts. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 5: Accounting with Tally

Company creation, ledgers, vouchers, inventory, GST and reports.

Key ideas

- Explain the purpose and essential vocabulary of accounting with tally.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates accounting with tally. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 6: DTP & Graphics

Page layout, typography, image preparation and print-ready documents.

Key ideas

- Explain the purpose and essential vocabulary of dtp & graphics.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates dtp & graphics. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 7: Web Fundamentals

HTML, CSS, responsive pages, domains, hosting and publishing.

Key ideas

- Explain the purpose and essential vocabulary of web fundamentals.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates web fundamentals. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 8: Programming Logic

Algorithms, flowcharts, variables, conditions, loops and functions.

Key ideas

- Explain the purpose and essential vocabulary of programming logic.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates programming logic. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 9: Internet & Cyber Safety

Email, cloud tools, privacy, passwords, phishing and digital citizenship.

Key ideas

- Explain the purpose and essential vocabulary of internet & cyber safety.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates internet & cyber safety. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 10: Capstone Project

Plan, build, test, document and present an office automation solution.

Key ideas

- Explain the purpose and essential vocabulary of capstone project.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates capstone project. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Practical Workbook

Complete these tasks in order. The instructor may adapt names and sample data, but every learner should produce original work.

Practical 1. Create and save a correctly named practice file for Computer Fundamentals. Include a short note describing what you learned.

Practical 2. Create and save a correctly named practice file for Office Productivity. Include a short note describing what you learned.

Practical 3. Create and save a correctly named practice file for Advanced Excel & MIS. Include a short note describing what you learned.

Practical 4. Create and save a correctly named practice file for Database Concepts. Include a short note describing what you learned.

Practical 5. Create and save a correctly named practice file for Accounting with Tally. Include a short note describing what you learned.

Practical 6. Create and save a correctly named practice file for DTP & Graphics. Include a short note describing what you learned.

Portfolio checklist

- Use folder format: CourseCode / StudentName / Module / Activity.
- Keep editable source files plus exported PDF or screenshot.
- Do not copy another student work; mention sources used.
- Back up important work in two locations.

Assignments & Assessment

Assignment 1: Concept Check

Write 300-500 words explaining three important concepts from Computer Fundamentals and Office Productivity. Add one real-life example.

Assignment 2: Practical Submission

Complete a combined task using Advanced Excel & MIS and Database Concepts. Submit source file, output and a one-page process note.

Assignment 3: Problem Solving

Identify a realistic institute or small-business problem that can be improved using Advanced Diploma in Computer Applications. Propose steps, tools, risks and success measures.

Assignment 4: Final Project

Build a complete project using at least four course modules. Test it, prepare screenshots, and present the result in 5-7 minutes.

Criterion	Weight
Accuracy and completeness	30%
Practical application	30%
Presentation and file organisation	20%
Originality, explanation and safe practice	20%

Revision & Safe Learning Guide

- Review class notes within 24 hours and practise again without watching the demonstration.
- Use legal software and trusted downloads. Scan external drives and downloaded files.
- Never share passwords, OTPs, private documents or student data.
- Verify AI-generated or internet information before using it in assignments.
- Use clear filenames, version numbers and regular backups.
- Ask for help when an error could damage data, equipment or accounts.

Quick viva questions

1. What is the purpose of Computer Fundamentals, and what common mistake should a beginner avoid?
2. What is the purpose of Office Productivity, and what common mistake should a beginner avoid?
3. What is the purpose of Advanced Excel & MIS, and what common mistake should a beginner avoid?
4. What is the purpose of Database Concepts, and what common mistake should a beginner avoid?
5. What is the purpose of Accounting with Tally, and what common mistake should a beginner avoid?
6. What is the purpose of DTP & Graphics, and what common mistake should a beginner avoid?

Support

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This foundation pack supports instruction; software screens and rules may change. Follow current instructor demonstrations and official documentation.